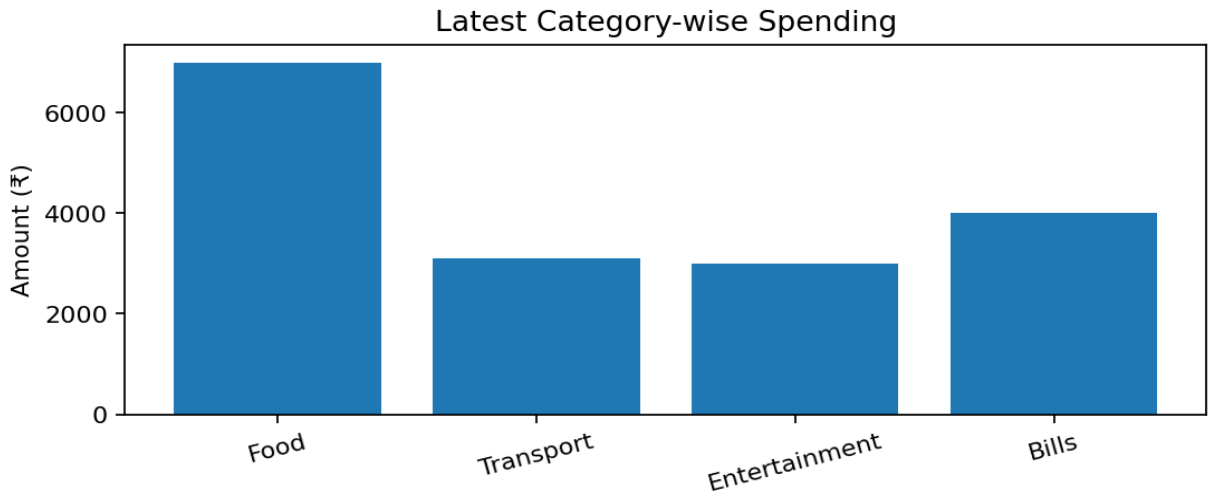
5.2 Expense Prediction

Spending Pattern Analysis

Spending Pattern Analysis examines historical transactions to discover repeated purchases, high-spending periods, and changes in category-wise expenses. By converting transaction records into useful patterns, the assistant can help users understand where their money is commonly spent and use this information for future budget planning.

Objectives

- Identify frequently used spending categories.
- Find categories with increasing expenses.
- Detect high-spending periods.
- Compare spending across months.
- Use historical patterns to improve planning.



Working Procedure

1. Collect transaction records.
2. Group transactions by category.
3. Calculate totals for each category.
4. Compare categories across different periods.
5. Identify increases, decreases, and repeated patterns.

Example Data Table

Category	June	July	August	Pattern
Food	■6,000	■6,500	■7,000	Increasing
Transport	■3,000	■3,200	■3,100	Mostly stable

Category	June	July	August	Pattern
Entertainment	■1,500	■2,500	■3,000	Increasing
Bills	■4,000	■4,000	■4,000	Recurring

Python Implementation

```
expenses = {
"Food": [6000, 6500, 7000],
"Transport": [3000, 3200, 3100],
"Entertainment": [1500, 2500, 3000],
"Bills": [4000, 4000, 4000]
}

for category, values in expenses.items():
change = values[-1] - values[0]
print(category, "Change:", change)

if change > 0:
print(" Pattern: spending increased")
elif change < 0:
print(" Pattern: spending decreased")
else:
print(" Pattern: stable")
```

Benefits

Pattern analysis helps users understand their financial behavior. Repeated increases in a category can be considered when setting future budgets.

Note: The values shown are illustrative examples for an academic project. A real application can use transaction records stored in a database.